

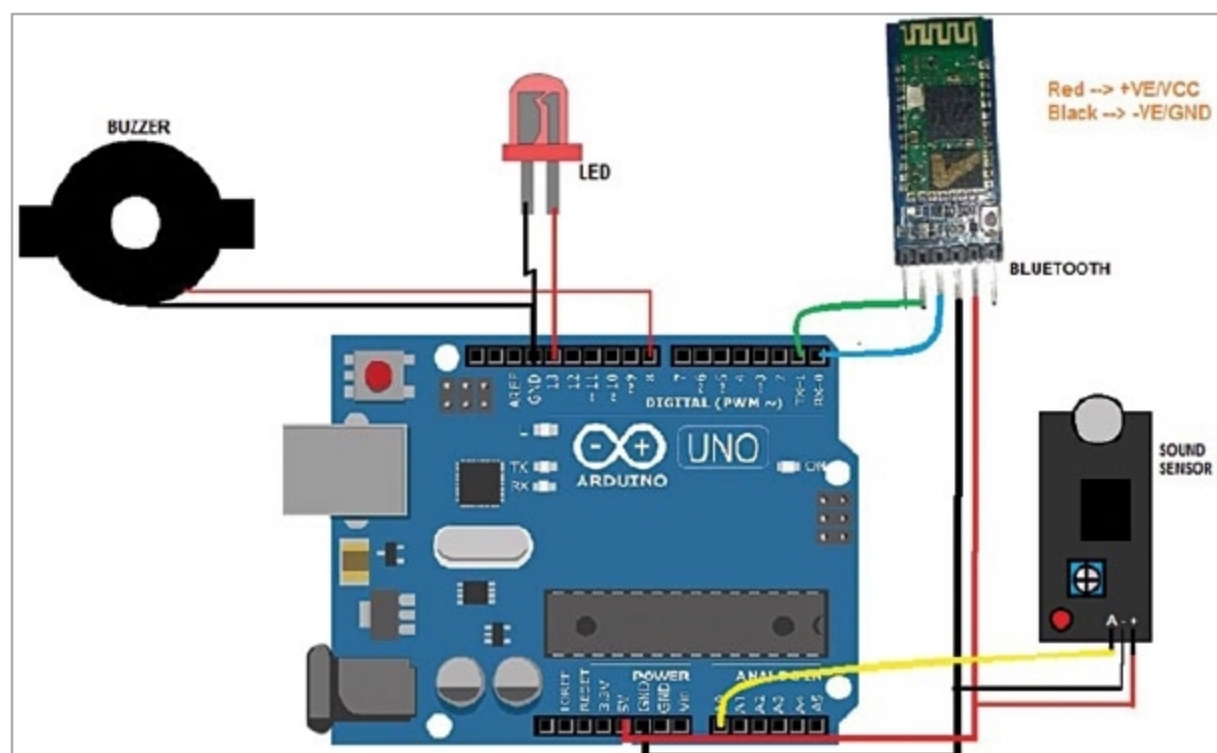


Fig. 7: Circuit diagram of the noise detector for silence zone

Arduino Pin Connections With Components

Arduino pins	Components
GND	BUZZER -ve
Pin 8	BUZZER +ve
Tx	Bluetooth Rx
Rx	Bluetooth Tx
5V	Bluetooth Vcc
GND	Bluetooth GND
Pin13	LED +ve
GND	LED -ve
5V	Sound sensor Vcc
GND	Sound sensor GND
A0	Sound sensor output

average output is given by sum of all readings/number of samples taken. In this project, we considered 160 samples of various noise levels to get the average output.

Building App

MIT App Inventor is used to develop the Android app for this project. Create a layout in the app, as shown in Fig. 5, using the following components:

- 2 text level
- 1 text input
- 1 list picker
- 1 button

To start building the app, open MIT

App Inventor from <https://appinventor.mit.edu> and set the code in Blocks Editor section, as shown in Fig. 6. Save the code as `silence_zone` and save its .apk file to the computer. This .apk file will be used in the Android phone later.

Circuit connections and testing

After developing the firmware and the app, connect the components with Arduino, as shown in Fig. 7. Arduino pin connection details are also shown in the table.

Power Arduino board using a 5V DC adaptor/power bank. Copy `silence_zone.apk` file from the computer to the Android phone. Install the app and open it from the phone. Tap on Bluetooth icon in the app and select HC-05 from the list to connect the phone with Bluetooth HC-05 in the circuit.

You will now be able to see the noise level value in the text bar in the app. When noise level reaches the threshold value, the LED light and buzzer will automatically get triggered by Arduino.

You can also reset the threshold value using the app. To do so, tap on the text box next to Send, enter the value you want to add and click on Send to set the threshold value. **EFY**

EFY Note

The source code of this project is available for free download at source.efymag.com

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Anyone who stops learning is old, whether at twenty or eighty.

—Henry Ford

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